Coding & Solution

Date	15 March 2026
Team ID	SWTID-2026-9550
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented ServiceNow solution. The project demonstrates script-controlled Access Control Lists (ACLs) that restrict record-level access based on the Branch field value and user roles.
Instructions:

1. Ensure the ACL configuration and scripts are well-structured and follow ServiceNow security best practices.

2. The solution addresses the core requirement of restricting access to Institution Details records based on the Branch value and assigned roles.

3. Include all ACL configurations, custom table fields, scripts, and setup steps needed to reproduce the application.

4. Attach or link the project repository if available.

Solution Summary

Field	Details
Repository Link / URL	Not provided in the project document
Programming Language(s)	ServiceNow scripting (script-based ACL configuration)
Framework(s) Used	ServiceNow platform
Key Features Implemented	Custom Institution Details table; Branch choices ECE/EEE/CSE; script-controlled READ ACL; role-based CREATE, WRITE and DELETE ACLs; admin full access; role-based record protection.
Pending / Incomplete Features	Repository URL and any additional implementation details not included in the project document.
Setup / Run Instructions	Create EEE User and roles bb1–bb4; create the Institution Details table and fields; create READ/CREATE/WRITE/DELETE ACLs; elevate security_admin; assign roles; impersonate users and verify access.

Core ACL Script

```
(function () {
// Allow admin users full access
if (gs.hasRole('admin')) {
return true;
}

// Allow users with bb1 role to see EEE records
if (gs.hasRole('bb1')) {
return true;
}

// Deny access for all others
return false;
})();
```

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	No
5	The application runs without critical errors	Yes*
6	Code is committed to a version control repository	No*

* Status is based only on the supplied project document: verification steps are documented, but no repository link is supplied.

Implementation Summary

Milestone	Implementation / Verification
1. Users & Roles	Created EEE User and custom roles bb1, bb2, bb3 and bb4; roles are assigned to the test user.
2. Table Creation	Created u_institution_details with Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description fields.
3. READ ACL	Record-level read ACL with bb1 and Branch = EEE condition; admin users receive full access.
4. CREATE ACL	Record-level create ACL requiring bb2.
5. WRITE ACL	Record-level write ACL requiring bb3.
6. DELETE ACL	Record-level delete ACL requiring bb4.

Additional Notes / Comments:

• The solution is designed to enforce record-level security in ServiceNow.

• The READ ACL uses the Branch value and bb1 role, while CREATE, WRITE and DELETE operations are controlled by bb2, bb3 and bb4 respectively.

• The project document states that the completed lab demonstrates protection against unauthorized viewing, modification, creation and deletion of records.

Conclusion

By completing this lab, learners gain an understanding of how script-controlled READ, WRITE, CREATE and DELETE ACLs enforce record-level security, and how user roles and record field values are evaluated together to make access decisions.